

Year 7 · Chapter 1 · Integers: answers (teacher copy)

1.01 Integers and opposites (booklet p. 3)

Step 1: What is an integer?

E1 (WE DO) Where is the dot?
the dot is at -2

1 Where is the dot?
the dot is at -4

2 Where is the dot?
the dot is at 3

3 Where is the dot?
the dot is at -7

4 Where is the dot?
the dot is at -1

5 Which integer is 6 left of 0?
it is at -6

Step 2: Integers in real life

E2 (WE DO) 3°C below zero
 $= -3$

6 a gain of 4 kg
 $= 4$

7 8 m below sea level
 $= -8$

8 a loss of \$12
 $= -12$

9 6 floors down
 $= -6$

10 a rise of 15 cm
 $= 15$

11 spending \$30
 $= -30$

Step 3: Opposites

E3 (WE DO) 7
opposite: -7

12 -5
opposite: 5

13 9
opposite: -9

14 -12
opposite: 12

15 30
opposite: -30

16 0
opposite: 0

17 -100
opposite: 100

Step 4: Double opposites

E4 (WE DO) $-(-3)$
 $= \text{opposite of } -3 \cdot = 3$

E5 (WE DO) $-(+4)$
 $= \text{opposite of } 4 \cdot = -4$

18 $-(-8)$
 $= \text{opposite of } -8 \cdot = 8$

19 $-(+5)$
 $= \text{opposite of } 5 \cdot = -5$

20 $-(-20)$
 $= \text{opposite of } -20 \cdot = 20$

21 $-(+12)$
 $= \text{opposite of } 12 \cdot = -12$

22 $-(-45)$
 $= \text{opposite of } -45 \cdot = 45$

23 $-(+30)$
 $= \text{opposite of } 30 \cdot = -30$

Step 5: Distance from zero

E6 (WE DO) -4 or 3
distances: 4 and $3 \cdot$ further: -4

24 -8 or 5
distances: 8 and $5 \cdot$ further: -8

25 6 or -2
distances: 6 and $2 \cdot$ further: 6

26 -10 or 9
distances: 10 and $9 \cdot$ further: -10

27 -3 or -11
distances: 3 and $11 \cdot$ further: -11

28 14 or -15
distances: 14 and $15 \cdot$ further: -15

Step 6: Is it right?

E7 (WE DO) Tom says $-(-4) = -4$. Is he right?
 $= \text{opposite of } -4 \cdot = 4 \cdot$ No. The answer is 4 .

29 Ali says $-(+7) = 7$. Is he right?
 $= \text{opposite of } 7 \cdot = -7 \cdot$ No. The answer is -7 .

30 Ben says -6 is further from 0 than 4. Is he right?
distances: 6 and $4 \cdot$ 6 is more than $4 \cdot$ Yes. Ben is right.

31 Mia says the opposite of 0 is 0. Is she right?
 0 is 0 steps from $0 \cdot$ its opposite: $0 \cdot$ Yes. Mia is right.

32 Jo says 10 m down and 10 m up are opposites. Is Jo right?
down: -10 , up: $10 \cdot$ same distance, other side $\cdot$ Yes. Jo is right.

1.02 The number line (booklet p. 5)

Step 1: Reading a number line

E1 (WE DO) Where is P?
P is at -5

1 Where is Q?

Q is at -1

2 Where is R?

R is at 3

3 Where is S?

S is at -7

4 Where is T?

T is at 5

5 Which integer is between -8 and -6?

it is -7

Step 2: Moving left and right

E2 (WE DO) Start at 3. Move 5 left.

land on -2

6 Start at -2. Move 6 right.

land on 4

7 Start at 1. Move 4 left.

land on -3

8 Start at -3. Move 5 left.

land on -8

9 Start at -7. Move 10 right.

land on 3

10 Start at 6. Move 13 left.

land on -7

Step 3: Distance between

E3 (WE DO) -2 and 4

2 to 0, 4 from 0 · = 6

E4 (WE DO) -7 and -3

count -7 to -3 · = 4

11 -1 and 6

1 to 0, 6 from 0 · = 7

12 -5 and 3

5 to 0, 3 from 0 · = 8

13 -9 and -2

count -9 to -2 · = 7

14 4 and -8

8 to 0, 4 from 0 · = 12

15 -12 and -5

count -12 to -5 · = 7

16 -15 and 10

15 to 0, 10 from 0 · = 25

Step 4: Scales in 2s, 5s and 10s

E5 (WE DO) What is at the ?

$10 \div 5 = 2$, step 2 · ? = 8

17 What is at the ?

$8 \div 4 = 2$, step 2 · ? = 4

18 What is at the ?

$20 \div 4 = 5$, step 5 · ? = 10

19 What is at the ?

$20 \div 2 = 10$, step 10 · ? = 10

20 The scale is -15, -10, □, 0. What is □?

-15 to -10 is 5, step 5 · □ = -5

Step 5: Halfway between

E6 (WE DO) -4 and 2

6 apart, half is 3 · $-4 + 3 = -1$

21 -6 and 2

8 apart, half is 4 · $-6 + 4 = -2$

22 -10 and 0

10 apart, half is 5 · $-10 + 5 = -5$

23 -3 and 9

12 apart, half is 6 · $-3 + 6 = 3$

24 -11 and -1

10 apart, half is 5 · $-11 + 5 = -6$

25 -15 and 5

20 apart, half is 10 · $-15 + 10 = -5$

Step 6: Number line problems

E7 (WE DO) An ant starts at 3. It walks 7 left. Where is it now?

start at 3, move 7 left · land on -4

26 A frog starts at -4. It hops 9 right. Where is it now?

start at -4, move 9 right · land on 5

27 Point A is at -6. Point B is 10 to the right of A. Where is B?

start at -6, move 10 right · land on 4

28 An ant at 5 walks 8 left, then 2 right. Where is it now?

start at 5, move 8 left: -3 · move 2 right · land on -1

29 A lift at level -3 goes up 7 floors, then down 6. Which level is it on?

start at -3, move 7 up: 4 · move 6 down · land on -2

30 (Check yourself) Which integer is halfway between -9 and 3?

= -3

1.03 Comparing and ordering integers

(booklet p. 7)

Step 1: Which is larger?

E1 (WE DO) -3 or 2

larger: 2

1 -6 or -2

larger: -2

2 1 or -4

larger: 1

$$3 \quad -8 \text{ or } -3$$

larger: -3

$$4 \quad -1 \text{ or } -7$$

larger: -1

$$5 \quad -12 \text{ or } -9$$

larger: -9

Step 2: Using < and >

$$\text{E2 (WE DO)} -4 \square 3$$

larger: 3 · symbol: <

$$6 \quad -2 \square -6$$

larger: -2 · symbol: >

$$7 \quad 0 \square -5$$

larger: 0 · symbol: >

$$8 \quad -9 \square -3$$

larger: -3 · symbol: <

$$9 \quad 4 \square -10$$

larger: 4 · symbol: >

$$10 \quad -15 \square -14$$

larger: -14 · symbol: <

$$11 \quad -20 \square 1$$

larger: 1 · symbol: <

Step 3: Ordering

$$\text{E3 (WE DO)} \text{ Ascending: } 3, -4, 0, -1$$

smallest: -4 · -4, -1, 0, 3

$$\text{E4 (WE DO)} \text{ Descending: } -2, 5, -7, 1$$

largest: 5 · 5, 1, -2, -7

$$12 \quad \text{Ascending: } 2, -3, -1, 4$$

smallest: -3 · -3, -1, 2, 4

$$13 \quad \text{Descending: } -4, 6, -1, 0$$

largest: 6 · 6, 0, -1, -4

$$14 \quad \text{Ascending: } -12, 8, -15, -3, 1$$

smallest: -15 · -15, -12, -3, 1, 8

$$15 \quad \text{Descending: } -20, -2, -11, 5, -8$$

largest: 5 · 5, -2, -8, -11, -20

Step 4: Comparing temperatures

$$\text{E5 (WE DO)} \text{ Which is colder: } -4^{\circ}\text{C or } -9^{\circ}\text{C?}$$

-9 is lower than -4 · colder: -9°C

$$16 \quad \text{Which is colder: } -2^{\circ}\text{C or } -6^{\circ}\text{C?}$$

-6 is lower than -2 · colder: -6°C

$$17 \quad \text{Which is colder: } 3^{\circ}\text{C or } -1^{\circ}\text{C?}$$

-1 is lower than 3 · colder: -1°C

$$18 \quad \text{Which is warmer: } -10^{\circ}\text{C or } -7^{\circ}\text{C?}$$

-7 is higher than -10 · warmer: -7°C

$$19 \quad \text{Which is colder: } -21^{\circ}\text{C or } -12^{\circ}\text{C?}$$

-21 is lower than -12 · colder: -21°C

Step 5: Simplify, then compare

$$\text{E6 (WE DO)} -(-3) \square -4$$

= 3 □ -4 · symbol: >

$$20 \quad -(-1) \square -2$$

= 1 □ -2 · symbol: >

$$21 \quad -(+4) \square -3$$

= -4 □ -3 · symbol: <

$$22 \quad -(-5) \square 6$$

= 5 □ 6 · symbol: <

$$23 \quad -7 \square -(+9)$$

= -7 □ -9 · symbol: >

$$24 \quad -(+6) \square 5$$

= -6 □ 5 · symbol: <

$$25 \quad -8 \square -(-8)$$

= -8 □ 8 · symbol: <

Step 6: Is it right?

$$\text{E7 (WE DO)} \text{ Tom says } -9 \text{ is larger than } -4. \text{ Is he right?}$$

-9 is left of -4 · so -9 is smaller · No. -4 is larger.

$$26 \quad \text{Mia says } -1 > -10. \text{ Is she right?}$$

-1 is right of -10 · so -1 is larger · Yes. Mia is right.

$$27 \quad \text{Ben says } -20^{\circ}\text{C is colder than } -2^{\circ}\text{C. Is he right?}$$

-20 is lower than -2 · so -20°C is colder · Yes. Ben is right.

$$28 \quad \text{Ali says } -5, -7, 0, 2 \text{ is in ascending order. Is he right?}$$

smallest: -7 · -7, -5, 0, 2 · No. It is -7, -5, 0, 2.

$$29 \quad \text{Jo says } -(-6) < -6. \text{ Is Jo right?}$$

-(-6) = 6 · 6 is right of -6 · No. -(-6) > -6.

1.04 Adding integers (booklet p. 9)

Step 1: Adding a positive

$$\text{E1 (WE DO)} -2 + 4$$

= 2

$$1 \quad -4 + 3$$

= -1

$$2 \quad -5 + 2$$

= -3

$$3 \quad -6 + 4$$

= -2

$$4 \quad -3 + 7$$

= 4

$$5 \quad -7 + 7$$

= 0

Step 2: Adding a negative

$$\text{E2 (WE DO)} 6 + (-2)$$

= 6 - 2 · = 4

$$\text{E3 (WE DO)} -2 + (-3)$$

= -2 - 3 · = -5

$$6 \quad 7 + (-3)$$

= 7 - 3 · = 4

$$7 \quad 5 + (-5)$$

= 5 - 5 · = 0

$$8 \quad 2 + (-6)$$

= 2 - 6 · = -4

$$9 \quad 4 + (-10) \\ = 4 - 10 \quad \cdot \quad = -6$$

$$10 \quad -1 + (-4) \\ = -1 - 4 \quad \cdot \quad = -5$$

$$11 \quad -3 + (-6) \\ = -3 - 6 \quad \cdot \quad = -9$$

Step 3: Zero pairs

$$E4 \text{ (WE DO)} -5 + 8 \\ 8 - 5 = 3 \quad \cdot \quad + \text{ has more: } 3$$

$$E5 \text{ (WE DO)} 6 + (-9) \\ 9 - 6 = 3 \quad \cdot \quad - \text{ has more: } -3$$

$$12 \quad -7 + 10 \\ 10 - 7 = 3 \quad \cdot \quad + \text{ has more: } 3$$

$$13 \quad -9 + 4 \\ 9 - 4 = 5 \quad \cdot \quad - \text{ has more: } -5$$

$$14 \quad 8 + (-12) \\ 12 - 8 = 4 \quad \cdot \quad - \text{ has more: } -4$$

$$15 \quad -15 + 20 \\ 20 - 15 = 5 \quad \cdot \quad + \text{ has more: } 5$$

$$16 \quad 25 + (-40) \\ 40 - 25 = 15 \quad \cdot \quad - \text{ has more: } -15$$

$$17 \quad -36 + 50 \\ 50 - 36 = 14 \quad \cdot \quad + \text{ has more: } 14$$

Step 4: Three numbers

$$E6 \text{ (WE DO)} 1 + (-4) + 5 \\ = -3 + 5 \quad \cdot \quad = 2$$

$$18 \quad 2 + (-5) + 4 \\ = -3 + 4 \quad \cdot \quad = 1$$

$$19 \quad -3 + 6 + (-2) \\ = 3 + (-2) \quad \cdot \quad = 1$$

$$20 \quad 12 + (-15) + (-6) \\ = -3 + (-6) \quad \cdot \quad = -9$$

$$21 \quad -20 + 8 + 15 \\ = -12 + 15 \quad \cdot \quad = 3$$

Step 5: Adding in real life

$$E7 \text{ (WE DO)} \text{ It is } -5^{\circ}\text{C}. \text{ The temperature rises } 8^{\circ}\text{C}. \text{ What is the temperature now?} \\ -5 + 8 \quad \cdot \quad = 3^{\circ}\text{C}$$

$$22 \quad \text{It is } -4^{\circ}\text{C} \text{ at dawn. By lunch it has warmed up } 9^{\circ}\text{C}. \text{ What is the temperature at lunch?} \\ -4 + 9 \quad \cdot \quad = 5^{\circ}\text{C}$$

$$23 \quad \text{A lift is at level } -2. \text{ It goes up } 5 \text{ floors. Which level is it on now?} \\ -2 + 5 \quad \cdot \quad = \text{level } 3$$

$$24 \quad \text{Sam has } \$15 \text{ in the bank. Sam spends } \$40 \text{ using credit. What is the balance now?} \\ 15 + (-40) \quad \cdot \quad = -\$25$$

$$25 \quad \text{A diver is at } -18 \text{ m. She swims up } 7 \text{ m, then down } 4 \text{ m. Where is she now?} \\ -18 + 7 + (-4) \quad \cdot \quad = -11 + (-4) \quad \cdot \quad = -15 \text{ m}$$

Step 6: Is it right?

$$E8 \text{ (WE DO)} \text{ Tom says } -6 + 2 = -8. \text{ Is he right?} \\ \text{start at } -6, \text{ move } 2 \text{ right } \quad \cdot \quad = -4 \quad \cdot \quad \text{No. The answer is } -4.$$

$$26 \quad \text{Mia says } -8 + 5 = -13. \text{ Is she right?} \\ \text{start at } -8, \text{ move } 5 \text{ right } \quad \cdot \quad = -3 \quad \cdot \quad \text{No. The answer is } -3.$$

$$27 \quad \text{Ben says } -7 + 3 + (-3) = -7. \text{ Is he right?} \\ 3 + (-3) \text{ is a zero pair } \quad \cdot \quad -7 + 0 = -7 \quad \cdot \quad \text{Yes. Ben is right.}$$

$$28 \quad \text{Ali says } 6 + (-9) = 3. \text{ Is he right?} \\ 9 - 6 = 3 \quad \cdot \quad - \text{ has more: } -3 \quad \cdot \quad \text{No. The answer is } -3.$$

$$29 \quad \text{Jo says adding always makes a number bigger. Is Jo right?} \\ 4 + (-6) = -2 \quad \cdot \quad -2 \text{ is smaller than } 4 \quad \cdot \quad \text{No. It can get smaller.}$$

1.05 Subtracting integers (booklet p. 11)

Step 1: Subtracting a positive

$$E1 \text{ (WE DO)} 1 - 4 \\ = -3$$

$$1 \quad 3 - 5 \\ = -2$$

$$2 \quad 0 - 6 \\ = -6$$

$$3 \quad 2 - 7 \\ = -5$$

$$4 \quad 4 - 12 \\ = -8$$

$$5 \quad 5 - 11 \\ = -6$$

Step 2: Starting below zero

$$E2 \text{ (WE DO)} -2 - 3 \\ \text{move } 3 \text{ left } \quad \cdot \quad = -5$$

$$6 \quad -1 - 4 \\ \text{move } 4 \text{ left } \quad \cdot \quad = -5$$

$$7 \quad -5 - 2 \\ \text{move } 2 \text{ left } \quad \cdot \quad = -7$$

$$8 \quad -6 - 6 \\ \text{move } 6 \text{ left } \quad \cdot \quad = -12$$

$$9 \quad -10 - 5 \\ \text{move } 5 \text{ left } \quad \cdot \quad = -15$$

$$10 \quad -8 - 12 \\ \text{move } 12 \text{ left } \quad \cdot \quad = -20$$

$$11 \quad -25 - 15 \\ \text{move } 15 \text{ left } \quad \cdot \quad = -40$$

Step 3: Subtracting a negative

$$E3 \text{ (WE DO)} 4 - (-2) \\ = 4 + 2 \quad \cdot \quad = 6$$

$$12 \quad 6 - (-3) \\ = 6 + 3 \quad \cdot \quad = 9$$

$$13 \quad 1 - (-7) \\ = 1 + 7 \quad \cdot \quad = 8$$

$$14 \quad -2 - (-5) \\ = -2 + 5 \quad \cdot = 3$$

$$15 \quad -8 - (-3) \\ = -8 + 3 \quad \cdot = -5$$

$$16 \quad -4 - (-4) \\ = -4 + 4 \quad \cdot = 0$$

$$17 \quad -15 - (-20) \\ = -15 + 20 \quad \cdot = 5$$

Step 4: Mixed + and -

$$E4 \text{ (WE DO)} \quad 2 - (-3) - 6 \\ = 2 + 3 - 6 \quad \cdot = 5 - 6 \quad \cdot = -1$$

$$18 \quad 1 - (-4) - 7 \\ = 1 + 4 - 7 \quad \cdot = 5 - 7 \quad \cdot = -2$$

$$19 \quad -3 - 5 - (-2) \\ = -3 - 5 + 2 \quad \cdot = -8 + 2 \quad \cdot = -6$$

$$20 \quad -6 - (-9) + (-5) \\ = -6 + 9 + (-5) \quad \cdot = 3 + (-5) \quad \cdot = -2$$

$$21 \quad 10 - (-2) - 15 \\ = 10 + 2 - 15 \quad \cdot = 12 - 15 \quad \cdot = -3$$

Step 5: Change and difference

$$E5 \text{ (WE DO)} \quad \text{It was } -3^{\circ}\text{C, then } 5^{\circ}\text{C. How much did it rise?} \\ 5 - (-3) \quad \cdot = 5 + 3 \quad \cdot = 8^{\circ}\text{C}$$

$$22 \quad \text{It was } -6^{\circ}\text{C, then } 2^{\circ}\text{C. How much did it rise?} \\ 2 - (-6) \quad \cdot = 2 + 6 \quad \cdot = 8^{\circ}\text{C}$$

$$23 \quad \text{A gull is at 25 m. A diver is at } -15 \text{ m. How far apart are they?} \\ 25 - (-15) \quad \cdot = 25 + 15 \quad \cdot = 40 \text{ m}$$

$$24 \quad \text{A freezer is } -18^{\circ}\text{C. The kitchen is } 22^{\circ}\text{C. What is the difference?} \\ 22 - (-18) \quad \cdot = 22 + 18 \quad \cdot = 40^{\circ}\text{C}$$

$$25 \quad \text{It was } 4^{\circ}\text{C, then } -9^{\circ}\text{C. What was the change?} \\ -9 - 4 \quad \cdot = -13^{\circ}\text{C}$$

Step 6: Is it right?

$$E6 \text{ (WE DO)} \quad \text{Tom says } -4 - 3 = -1. \text{ Is he right?} \\ \text{start at } -4, \text{ move 3 left } \cdot = -7 \quad \cdot \text{ No. The answer is } -7.$$

$$26 \quad \text{Mia says } 2 - (-6) = -4. \text{ Is she right?} \\ = 2 + 6 \quad \cdot = 8 \quad \cdot \text{ No. The answer is 8.}$$

$$27 \quad \text{Ben says } -5 - (-5) = 0. \text{ Is he right?} \\ = -5 + 5 \quad \cdot = 0 \quad \cdot \text{ Yes. Ben is right.}$$

$$28 \quad \text{Ali says } -1 - 8 = -9. \text{ Is he right?} \\ \text{start at } -1, \text{ move 8 left } \cdot = -9 \quad \cdot \text{ Yes. Ali is right.}$$

$$29 \quad \text{Jo says subtracting always makes a number smaller. Is Jo right?} \\ 4 - (-3) = 7 \quad \cdot 7 \text{ is larger than 4 } \cdot \text{ No. It can get larger.}$$

1.06 Multiplying integers (booklet p. 13)

Step 1: Equal groups

$$E1 \text{ (WE DO)} \quad 2 \times (-4) \\ (-4) + (-4) \quad \cdot = -8$$

$$1 \quad 2 \times (-3) \\ (-3) + (-3) \quad \cdot = -6$$

$$2 \quad 3 \times (-4) \\ (-4) + (-4) + (-4) \quad \cdot = -12$$

$$3 \quad 4 \times (-2) \\ (-2) + (-2) + (-2) + (-2) \quad \cdot = -8$$

$$4 \quad 3 \times (-10) \\ (-10) + (-10) + (-10) \quad \cdot = -30$$

$$5 \quad 2 \times (-25) \\ (-25) + (-25) \quad \cdot = -50$$

Step 2: The sign rules

$$E2 \text{ (WE DO)} \quad -3 \times 4 \\ \text{different signs: } - \quad \cdot = -12$$

$$6 \quad 5 \times (-2) \\ \text{different signs: } - \quad \cdot = -10$$

$$7 \quad -6 \times 3 \\ \text{different signs: } - \quad \cdot = -18$$

$$8 \quad -4 \times (-2) \\ \text{same signs: } + \quad \cdot = 8$$

$$9 \quad 7 \times (-5) \\ \text{different signs: } - \quad \cdot = -35$$

$$10 \quad -9 \times (-3) \\ \text{same signs: } + \quad \cdot = 27$$

$$11 \quad -8 \times 6 \\ \text{different signs: } - \quad \cdot = -48$$

Step 3: Why $- \times -$ is $+$

$$E3 \text{ (WE DO)} \quad 1 \times (-5) = -5, 0 \times (-5) = 0, \dots \\ -1 \times (-5) = 5 \quad \cdot -2 \times (-5) = 10$$

$$12 \quad 1 \times (-4) = -4, 0 \times (-4) = 0, \dots \\ -1 \times (-4) = 4 \quad \cdot -2 \times (-4) = 8$$

$$13 \quad 1 \times (-2) = -2, 0 \times (-2) = 0, \dots \\ -1 \times (-2) = 2 \quad \cdot -2 \times (-2) = 4$$

$$E4 \text{ (WE DO)} \quad -2 \times (-6) \\ \text{same signs: } + \quad \cdot = 12$$

$$14 \quad -5 \times (-4) \\ \text{same signs: } + \quad \cdot = 20$$

$$15 \quad -7 \times (-8) \\ \text{same signs: } + \quad \cdot = 56$$

Step 4: Three numbers

E5 (WE DO) $2 \times (-3) \times 4$
 $= -6 \times 4 \cdot = -24$

16 $-2 \times 5 \times 3$
 $= -10 \times 3 \cdot = -30$

17 $3 \times (-2) \times (-5)$
 $= -6 \times (-5) \cdot = 30$

18 $-3 \times 4 \times (-5)$
 $= -12 \times (-5) \cdot = 60$

19 $-4 \times (-2) \times (-3)$
 $= 8 \times (-3) \cdot = -24$

20 $-5 \times (-2) \times (-2) \times (-3)$
 $= 10 \times (-2) \times (-3) \cdot = -20 \times (-3) \cdot = 60$

Step 5: Multiplying in real life

E6 (WE DO) The temperature drops 2°C every hour for 4 hours. What is the total change?
 $4 \times (-2) \cdot = -8^{\circ}\text{C}$

21 A diver goes down 4 m every minute for 6 minutes. What is her change in height?
 $6 \times (-4) \cdot = -24 \text{ m}$

22 Sam spends \$5 a day for 7 days. What is the change in Sam's money?
 $7 \times (-5) \cdot = -\$35$

23 A share price falls \$3 a week for 9 weeks. What is the total change?
 $9 \times (-3) \cdot = -\$27$

24 A lift at level 3 goes down 2 floors, 5 times. Which level is it on now?
 $5 \times (-2) = -10 \cdot 3 + (-10) \cdot = \text{level } -7$

Step 6: Is it right?

E7 (WE DO) Tom says $-3 \times 6 = 18$. Is he right?
different signs: $- \cdot = -18$ · No. The answer is -18 .

25 Mia says $-7 \times (-2) = -14$. Is she right?
same signs: $+ \cdot = 14$ · No. The answer is 14.

26 Ben says $-4 \times \square = 28$ means $\square = -7$. Is he right?
 $-4 \times (-7) \cdot = 28 \cdot \text{Yes. Ben is right.}$

27 Ali says $3 \times (-2) \times (-5) = -30$. Is he right?
 $3 \times (-2) = -6 \cdot -6 \times (-5) = 30 \cdot \text{No. The answer is 30.}$

28 Jo says $-2 \times (-3) \times (-4)$ is positive. Is Jo right?
 $-2 \times (-3) = 6 \cdot 6 \times (-4) = -24 \cdot \text{No. It is } -24.$

1.07 Dividing integers (booklet p. 15)

Step 1: Division undoes $\times$

E1 (WE DO) $4 \times (-2) = -8$
 $-8 \div 4 = -2 \cdot -8 \div (-2) = 4$

1 $2 \times (-5) = -10$
 $-10 \div 2 = -5 \cdot -10 \div (-5) = 2$

2 $3 \times (-6) = -18$
 $-18 \div 3 = -6 \cdot -18 \div (-6) = 3$

3 $5 \times (-4) = -20$
 $-20 \div 5 = -4 \cdot -20 \div (-4) = 5$

4 $6 \times (-7) = -42$
 $-42 \div 6 = -7 \cdot -42 \div (-7) = 6$

Step 2: The sign rules

E2 (WE DO) $-18 \div 3$
different signs: $- \cdot = -3$

5 $-12 \div 4$
different signs: $- \cdot = -3$

6 $20 \div (-5)$
different signs: $- \cdot = -4$

7 $-30 \div (-6)$
same signs: $+ \cdot = 5$

8 $-42 \div 7$
different signs: $- \cdot = -6$

9 $-56 \div (-8)$
same signs: $+ \cdot = 7$

10 $72 \div (-9)$
different signs: $- \cdot = -8$

Step 3: Missing numbers

E3 (WE DO) $\square \div 2 = -4$
 $\square = -4 \times 2 \cdot \square = -8$

E4 (WE DO) $-20 \div \square = 5$
 $\square = -20 \div 5 \cdot \square = -4$

11 $\square \div 3 = -6$
 $\square = -6 \times 3 \cdot \square = -18$

12 $\square \div (-4) = 5$
 $\square = 5 \times (-4) \cdot \square = -20$

13 $-30 \div \square = -6$
 $\square = -30 \div (-6) \cdot \square = 5$

14 $\square \div (-7) = -3$
 $\square = -3 \times (-7) \cdot \square = 21$

15 $48 \div \square = -8$
 $\square = 48 \div (-8) \cdot \square = -6$

Step 4: Dividing step by step

E5 (WE DO) $-24 \div 2 \div 3$
 $= -12 \div 3 \cdot = -4$

16 $-40 \div 4 \div 2$
 $= -10 \div 2 \cdot = -5$

17 $36 \div (-3) \div 2$
 $= -12 \div 2 \cdot = -6$

18 $-48 \div (-4) \div (-3)$
 $= 12 \div (-3) \cdot = -4$

19 $-90 \div 3 \div (-5)$
 $= -30 \div (-5) \cdot = 6$

Step 5: Sharing equally

E6 (WE DO) A debt of \$20 is shared by 4 people. Write each share as an integer.
 $-20 \div 4 \cdot = -\$5 \text{ each}$

20 A debt of \$36 is shared by 3 people. Write each share as an integer.

$$-36 \div 3 = -\$12 \text{ each}$$

21 The temperature fell 15°C in 5 hours. What was the change each hour?

$$-15 \div 5 = -3^{\circ}\text{C each hour}$$

22 A diver went down 24 m in 6 minutes. What was her change each minute?

$$-24 \div 6 = -4 \text{ m each minute}$$

23 A share price fell \$56 over 8 days, the same each day. What was the change each day?

$$-56 \div 8 = -\$7 \text{ each day}$$

Step 6: Is it right?

E7 (WE DO) Tom says $-24 \div 6 = 4$. Is he right?

different signs: $- \div = -4$ · No. The answer is -4 .

24 Mia says $35 \div (-7) = -5$. Is she right?

different signs: $- \div = -5$ · Yes. Mia is right.

25 Ben says $-40 \div (-8) = -5$. Is he right?

same signs: $+ \div = 5$ · No. The answer is 5.

26 Ali says $\square \div (-2) = 6$ means $\square = -3$. Is he right?

$\square = 6 \times (-2)$ · $\square = -12$ · No. $\square$ is -12 , not -3 .

27 Jo says dividing by a negative always gives a negative. Is Jo right?

$-12 \div (-3) = 4$ · 4 is positive · No. It can be positive.

1.08 Order of operations with integers

(booklet p. 17)

Step 1: The order

E1 (WE DO) $-5 + 2 \times 3$

$$= -5 + 6 = 1$$

$$1 \quad -1 + 2 \times 5$$

$$= -1 + 10 = 9$$

$$2 \quad -10 + 3 \times 2$$

$$= -10 + 6 = -4$$

$$3 \quad 6 - 4 \times 3$$

$$= 6 - 12 = -6$$

$$4 \quad -8 + 12 \div 4$$

$$= -8 + 3 = -5$$

$$5 \quad -7 - 10 \div 2$$

$$= -7 - 5 = -12$$

$$6 \quad 4 \times (-3) + 5$$

$$= -12 + 5 = -7$$

Step 2: Brackets first

E2 (WE DO) $(-2 + 5) \times 3$

$$= 3 \times 3 = 9$$

$$7 \quad (-4 + 6) \times 5$$

$$= 2 \times 5 = 10$$

$$8 \quad (1 - 7) \times 2$$

$$= -6 \times 2 = -12$$

$$9 \quad (-3 - 2) \times 4$$

$$= -5 \times 4 = -20$$

$$10 \quad (-8 + 2) \div 3$$

$$= -6 \div 3 = -2$$

$$11 \quad -2 \times (5 - 9)$$

$$= -2 \times (-4) = 8$$

$$12 \quad (-9 - 6) \div 5$$

$$= -15 \div 5 = -3$$

Step 3: $\times$ and $\div$ before $+$ and $-$

E3 (WE DO) $2 \times (-3) + 12 \div 4$

$$= -6 + 3 = -3$$

$$13 \quad 3 \times (-2) + 8 \div 2$$

$$= -6 + 4 = -2$$

$$14 \quad -12 \div 3 - 2 \times 5$$

$$= -4 - 10 = -14$$

$$15 \quad 4 \times (-5) + (-18) \div 6$$

$$= -20 + (-3) = -23$$

$$16 \quad -3 \times (-4) - 20 \div (-5)$$

$$= 12 - (-4) = 12 + 4 = 16$$

Step 4: Left to right

E4 (WE DO) $-20 \div 5 \times 2$

$$= -4 \times 2 = -8$$

E5 (WE DO) $3 - 7 + 2$

$$= -4 + 2 = -2$$

$$17 \quad -18 \div 6 \times 3$$

$$= -3 \times 3 = -9$$

$$18 \quad 6 - 10 + 3$$

$$= -4 + 3 = -1$$

$$19 \quad 24 \div (-4) \times 2$$

$$= -6 \times 2 = -12$$

$$20 \quad -2 - 5 + 9$$

$$= -7 + 9 = 2$$

$$21 \quad -30 \div (-5) \times (-2)$$

$$= 6 \times (-2) = -12$$

Step 5: Putting it together

E6 (WE DO) $5 - (2 - 6) \times 3$

$$= 5 - (-4) \times 3 = 5 - (-12) = 17$$

$$22 \quad -4 + (1 - 3) \times 5$$

$$= -4 + (-2) \times 5 = -4 + (-10) = -14$$

$$23 \quad 10 - (-2 + 6) \times 3$$

$$= 10 - 4 \times 3 = 10 - 12 = -2$$

$$24 \quad (-8 + 2) \div 3 - 4$$

$$= -6 \div 3 - 4 = -2 - 4 = -6$$

$$25 \quad -3 \times (4 - 7) + (-10)$$

$$= -3 \times (-3) + (-10) = 9 + (-10) = -1$$

Step 6: Is it right?

E7 (WE DO) Tom says $-2 + 3 \times 4 = 4$. Is he right?

$3 \times 4 = 12$ first · $-2 + 12 = 10$ · No. The answer is 10.

26 Mia says $6 - 2 \times (-3) = 12$. Is she right?

$2 \times (-3) = -6$ first · $6 - (-6) = 12$ · Yes. Mia is right.

27 Ben says $-12 \div 2 \times 3 = -2$. Is he right?
left to right: $-12 \div 2 = -6$ · $-6 \times 3 = -18$ · **No. The answer is -18 .**

28 Ali says $(-5 + 3) \times 2 = -4$. Is he right?
brackets first: -2 · $-2 \times 2 = -4$ · **Yes. Ali is right.**

29 Jo says $-4 + 6 \times 3 = 6$. Is Jo right?
 6×3 first: $-4 + 18$ · $= 14$ · **No. $(-4 + 6) \times 3 = 6$.**

1.09 Integers on a calculator (booklet p. 19)

Step 1: The $(-)$ key

E1 (WE DO) $-48 + 25$
 $(-)$ $48 + 25 =$ · $= -23$

1 $-63 + 41$
 $(-)$ $63 + 41 =$ · $= -22$

2 $-27 - 38$
 $(-)$ $27 - 38 =$ · $= -65$

3 $-150 + 86$
 $(-)$ $150 + 86 =$ · $= -64$

4 $45 - 112$
 $45 - 112 =$ · $= -67$

5 $-208 - 97$
 $(-)$ $208 - 97 =$ · $= -305$

6 $-76 + 176$
 $(-)$ $76 + 176 =$ · $= 100$

Step 2: $\times$ and $\div$ with negatives

E2 (WE DO) -18×12
 $(-)$ $18 \times 12 =$ · $= -216$

7 -23×15
 $(-)$ $23 \times 15 =$ · $= -345$

8 $36 \times (-14)$
 $36 \times (-)$ $14 =$ · $= -504$

9 $-540 \div 12$
 $(-)$ $540 \div 12 =$ · $= -45$

10 $-27 \times (-31)$
 $(-)$ $27 \times (-)$ $31 =$ · $= 837$

11 $-1\,196 \div (-26)$
 $(-)$ $1196 \div (-)$ $26 =$ · $= 46$

Step 3: Check the sign first

E3 (WE DO) -34×29
different signs: $-$ · $= -986$

12 $-52 \times (-18)$
same signs: $+$ · $= 936$

13 $65 \times (-24)$
different signs: $-$ · $= -1\,560$

14 $-3\,024 \div 56$
different signs: $-$ · $= -54$

15 $-2\,491 \div (-47)$
same signs: $+$ · $= 53$

16 -125×16
different signs: $-$ · $= -2\,000$

Step 4: Brackets on the calculator

E4 (WE DO) $(-28 + 13) \times 6$
bracket: -15 · $= -90$

17 $(-45 - 19) \times 5$
bracket: -64 · $= -320$

18 $(36 - 81) \times (-4)$
bracket: -45 · $= 180$

19 $(-120 + 48) \div 8$
bracket: -72 · $= -9$

20 $(-67 - 58) \div (-25)$
bracket: -125 · $= 5$

Step 5: Estimate, then calculate

E5 (WE DO) -39×21
 $\approx -40 \times 20 = -800$ · $= -819$

21 -52×19
 $\approx -50 \times 20 = -1\,000$ · $= -988$

22 $-61 \times (-29)$
 $\approx -60 \times (-30) = 1\,800$ · $= 1\,769$

23 $78 \times (-41)$
 $\approx 80 \times (-40) = -3\,200$ · $= -3\,198$

24 $-403 + 198$
 $\approx -400 + 200 = -200$ · $= -205$

Step 6: Calculator problems

E6 (WE DO) An account has \$120. Four payments of \$38 come out. What is the balance?
 $120 - 4 \times 38$ · $= -\$32$

25 A submarine at -185 m rises 12 m a minute for 9 minutes. Where is it now?
 $-185 + 9 \times 12$ · $= -77$ m

26 It is -14°C . It warms 3°C an hour for 7 hours. What is the temperature?
 $-14 + 7 \times 3$ · $= 7^\circ\text{C}$

27 An account has \$85. Seven payments of \$26 come out. What is the balance?
 $85 - 7 \times 26$ · $= -\$97$

28 A diver at -42 m goes down 6 m a minute for 8 minutes. Where is she now?
 $-42 - 8 \times 6$ · $= -90$ m

29 (Check yourself) $(-37 - 89) \times (-15)$
 $= 1\,890$

1.10 Integer problems (booklet p. 21)

Step 1: Temperature

E1 (WE DO) -6°C , rises 4°C
 $-6 + 4$ · $= -2^\circ\text{C}$

1 -3°C , rises 9°C
 $-3 + 9$ · $= 6^\circ\text{C}$

2 2°C , falls 5°C
 $2 - 5$ · $= -3^\circ\text{C}$

3 -1°C , falls 6°C
 $-1 - 6$ · $= -7^\circ\text{C}$

4 -8°C , rises 8°C
 $-8 + 8 = 0^{\circ}\text{C}$

5 -12°C , rises 15°C
 $-12 + 15 = 3^{\circ}\text{C}$

Step 2: Above and below sea level

E2 (WE DO) A gull at 20 m and a diver at -5 m
 $20 - (-5) = 25$ m

6 A boat at 0 m and a diver at -12 m
 $0 - (-12) = 12$ m

7 A cliff top at 30 m and a diver at -8 m
 $30 - (-8) = 38$ m

8 A lighthouse top at 45 m and a diver at -20 m
 $45 - (-20) = 65$ m

9 A diver at -5 m and a wreck at -22 m
 $-5 - (-22) = 17$ m

10 A bird at 42 m and a shark at -37 m
 $42 - (-37) = 79$ m

Step 3: Lifts and floors

E3 (WE DO) A lift at level -1 goes up 4 floors. Which level is it on?
 $-1 + 4 = \text{level } 3$

11 A lift at level 2 goes down 5 floors. Which level is it on?
 $2 - 5 = \text{level } -3$

12 A lift at level -3 goes up 8 floors. Which level is it on?
 $-3 + 8 = \text{level } 5$

13 A lift at level 5 goes down 9 floors. Which level is it on?
 $5 - 9 = \text{level } -4$

14 A lift at level -2 goes down 2, then up 7. Which level is it on?
 $-2 - 2 + 7 = \text{level } 3$

Step 4: Money

E4 (WE DO) The balance is \$30. You pay \$45.
 $30 - 45 = -\$15$

15 The balance is \$12. You pay \$25.
 $12 - 25 = -\$13$

16 The balance is $-\$20$. You pay in \$50.
 $-20 + 50 = \$30$

17 The balance is $-\$15$. You pay \$30.
 $-15 - 30 = -\$45$

18 The balance is \$40. You pay \$25, then \$30.
 $40 - 25 - 30 = -\$15$

19 The balance is $-\$60$. You pay in \$35, then \$40.
 $-60 + 35 + 40 = \$15$

Step 5: Several steps

E5 (WE DO) It is -6°C . It warms 2°C an hour for 5 hours. How warm is it now?
 $2 \times 5 = 10, -6 + 10 = 4^{\circ}\text{C}$

20 A diver at -30 m rises 4 m a minute for 6 minutes. Where is she?
 $4 \times 6 = 24, -30 + 24 = -6$ m

21 It is 5°C . It cools 3°C an hour for 4 hours. How cold is it now?
 $4 \times (-3) = -12, 5 + (-12) = -7^{\circ}\text{C}$

22 A win scores $+3$, a loss -2 . A team wins 4 and loses 7. What is its score?
 $4 \times 3 + 7 \times (-2) = 12 + (-14) = -2$

23 Sam has \$50 and pays \$12 a week for 6 weeks. What is the balance?
 $6 \times 12 = 72, 50 - 72 = -\22

Step 6: Is it right?

E6 (WE DO) Tom says level -3 , up 5 floors, is level -8 . Is he right?
 $-3 + 5 = 2$ · No. It is at level 2.

24 Mia says 12 m and -9 m are 3 m apart. Is she right?
 $12 - (-9) = 21$ · No. They are 21 m apart.

25 Ben says $-\$20$, then \$35 in, leaves \$15. Is he right?
 $-20 + 35 = 15$ · Yes. Ben is right.

26 Ali says 3°C , falling 8°C , gives -5°C . Is he right?
 $3 - 8 = -5$ · Yes. Ali is right.

27 Jo says it is warmer at -4°C than at -9°C . Is Jo right?
 -4 is higher than -9 · so -4°C is warmer · Yes. Jo is right.